935K Methylation Chip: Frequently Asked Questions (FAQ)

Q1: How to determine whether the 935K chip includes CpG sites of interest?

A: The CpG probes on this chip comprehensively cover **CpG islands**, **promoter regions**, **enhancer regions**, and **gene coding regions**. You can assess coverage based on the chip's probe distribution. Additionally, we provide detailed probe location information to help verify whether specific sites are included.

Q2: What are the advantages of the 935K chip compared to the 850K chip?

A: The 935K chip maximizes compatibility with the 850K chip while improving performance:

- •Enhancers and super-enhancers identified in cancer studies.
- •CTCF-binding domains.
- •Removed 107,000 poorly performing CpG probes from the 850K chip.
- •Added > 186,000 new probes targeting.
- •Chromatin-open regions detected via **ATAC-seq** and **ChIP-seq** in primary tumors.
- •Expanded converage of CpG islands and improved exon coverage for accurate CNV detection.
- •Highercost-effectiveness with broader genomic coverage.

Q3: Why can't some methylation sites on the chip be annotated to genes? Weren't the sites designed for genes?

A: The chip's probes cover the **entire genome**, including regions beyond coding genes, such as:

- •CpG islands.
- •RefSeq gene regions.
- •ENCODE open chromatin regions.
- •FANTOM5 enhancer regions.
- •ENCODE open chromatin regions.
- •ENCODE transcription factor binding sites.

•miRNA promoter regions.

Some probes may map to **intergenic regions** or non-coding regions, which are not linked to specific genes.

Q4: Does methylation differential analysis support samples without biological replicates? What are the screening criteria?

A: No, biological replicates are typically required for differential analysis.

Differential analysis screening thresholds:

- **ΔBeta > 0.2** (absolute value).
- **Adjusted *P*-value (adjP) < 0.05**.

Thresholds may be adjusted based on project-specific requirements.

Q5: How to interpret results comparing control vs. treatment groups in differential analysis?

A: Results are labeled as ”control_vs_case”.

ΔBeta: Represents the methylation change in the case group relative to the control.

- **ΔBeta > 0**: Higher methylation in the case group (hypermethylation).
- **ΔBeta < 0**: Lower methylation in the case group (hypomethylation).

Q6: Why is SeSAMe software used for data analysis, instead of using GenomeStudio Methylation Module software?

A: There are lots of advantages by using SeSAMe software over GenomeStudio Methylation Module software. **1)** SeSAMe is command line/R-based that is more suitable for advanced users; **2)** SeSAMe employs superior normalization method for calculating Beta values, including Noob, pOOBAH, and Probe-Type Bias Normalization; **3)** While GenomeStudio provides visual QC, SeSAMe offers a more quantitative, automated, and stringent QC framework, like Sample-Level QC, Probe-Level QC, and identification of Cross-Reactive and Problematic Probes; **4)** Script-based workflows ensure full reproducibility and facilitate processing of large sample sets on high-performance computing clusters, unlike GenomeStudio's manual, GUI-limited approach. GenomeStudio can become slow and unstable with large sample sizes； **5)** The output of SeSAMe is an RGChannelSet or GenomicRatioSet object, which is the standard data structure for methylation analysis in

Bioconductor. This allows for immediate and effortless integration with a vast array of powerful R packages for: Differential Methylation Analysis, DMR identification, cell type deconvolution, and multi-omics integration.

In summary, while GenomeStudio suits initial data visualization, SeSAMe is superior for rigorous analysis, offering advanced processing, automated QC, reproducibility, and integration capabilities that make it the standard for publication-quality epigenome-wide studies.

*For additional technical details or custom analysis, please contact our support team.